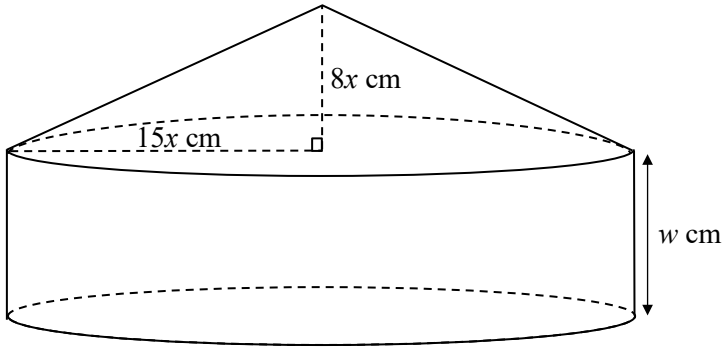
RAFFLES INSTITUTION
H2 Mathematics 9758
2026 Year 6 Term 3 Revision 1 (Lambda)
Topic(s): Calculus I
Differentiation, Applications, Maclaurin and Graphing (Solutions)

1 NYJC Prelim 9758/2022/02/Q3(modified)

[It is given that a right circular cone with base radius r and height h has volume $\frac{1}{3}\pi r^2 h$ and curved surface area πrl, where l is the slanted height of the cone.]



The model of a house is made up of three parts.

- The roof is modelled by the curved surface of a right circular cone with base radius $15x$ cm and height $8x$ cm.
- The walls are modelled by the curved surface of a cylinder of radius $15x$ cm and height w cm.
- The floor is modelled by a circular disc of radius $15x$ cm.

The three parts are joined together as shown in the diagram. The model is made of material of negligible thickness. It is given that the volume of the model is a fixed value k cm^3. Use differentiation to find the exact values of x and w in terms of k such that the external surface area is a minimum. Simplify your answers. [8]

Solution:

Let the external surface area be A.

$$A = 2\pi(15x)w + \pi(15x)^2 + \pi(15x)\sqrt{(8x)^2 + (15x)^2}$$

$$= 2\pi(15x)w + \pi(15x)^2 + \pi(15x)(17x)$$

$$= 30\pi xw + 480\pi x^2 \quad \text{----(1)}$$

$$\text{Volume} = k = \pi(15x)^2 w + \frac{1}{3}\pi(15x)^2(8x)$$

Note: k is a constant

$$k = 225\pi x^2 w + 600\pi x^3 \text{ ---(2)}$$

From (2), $w = \dfrac{k - 600\pi x^3}{225\pi x^2}$

Substitute into (1), $A = 30\pi x\left[\dfrac{k-600\pi x^3}{225\pi x^2}\right] + 480\pi x^2 = \dfrac{2k}{15x} + 400\pi x^2$.

$\dfrac{\mathrm{d}A}{\mathrm{d}x} = -\dfrac{2k}{15x^2} + 800\pi x$

When $\dfrac{\mathrm{d}A}{\mathrm{d}x} = 0$, $x^3 = \dfrac{k}{6000\pi} \Rightarrow x = \dfrac{1}{10}\left(\dfrac{k}{6\pi}\right)^{\frac{1}{3}}$

$\therefore w = \dfrac{1}{225\pi\dfrac{k^{\frac{2}{3}}}{100(6^{\frac{2}{3}})(\pi^{\frac{2}{3}})}}\left[k - 600\pi \times \dfrac{k}{6000\pi}\right]$

$= \dfrac{100(6^{\frac{2}{3}})}{225\pi^{\frac{1}{3}}k^{\frac{2}{3}}}\left[\dfrac{9}{10}k\right] = \dfrac{12}{5}\left(\dfrac{k}{6\pi}\right)^{\frac{1}{3}}$

Now, $\dfrac{\mathrm{d}^2A}{\mathrm{d}x^2} = \dfrac{4k}{15x^3} + 800\pi > 0$ since $k, x > 0$

Hence A is a minimum.

Note:

For students who use the 1st derivative test, you need to justify the sign of

$\dfrac{\mathrm{d}A}{\mathrm{d}x} = -\dfrac{2k}{15x^2} + 800\pi x$ clearly.

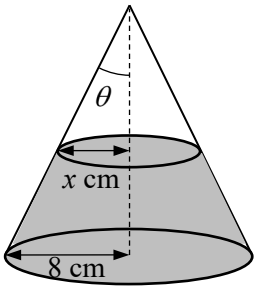
$\dfrac{\mathrm{d}A}{\mathrm{d}x} = -\dfrac{2k}{15x^2} + 800\pi x = \dfrac{12000\pi\left(x^3 - \dfrac{k}{6000\pi}\right)}{15x^2}$.

Since $\dfrac{12000\pi}{15x^2} > 0$ for all x, the sign of $\dfrac{\mathrm{d}A}{\mathrm{d}x}$ depends only on that of $\left(x^3 - \dfrac{k}{6000\pi}\right)$.

x	$\left(\dfrac{1}{10}\left(\dfrac{k}{6\pi}\right)^{\frac{1}{3}}\right)^-$	$\dfrac{1}{10}\left(\dfrac{k}{6\pi}\right)^{\frac{1}{3}}$	$\left(\dfrac{1}{10}\left(\dfrac{k}{6\pi}\right)^{\frac{1}{3}}\right)^+$
$\left(x^3 - \dfrac{k}{6000\pi}\right)$	< 0	0	> 0
$\dfrac{\mathrm{d}A}{\mathrm{d}x}$	< 0	0	> 0
Shape	\	___	/

Hence A is a minimum.

2 VJC Prelim 9758/2021/01/Q4



Water is poured at a rate of 8 cm^3 per second into a conical container with base radius 8 cm. The semi-vertical angle of the cone is θ, where $\tan\theta = 0.4$. At time t seconds after the start, the radius of the water surface is x cm (see diagram). Find the rate of increase of the depth of water when the depth is 10 cm.

[The volume of a cone of base radius r and height h is given by $V = \frac{1}{3}\pi r^2 h$.] [7]

Solution:

Let the volume of water in the cone be V cm^3 at time t seconds, when the depth is y cm.
Let h cm be the height of the container.

$\tan\theta = 0.4 = \frac{8}{h} \Rightarrow h = 20$

$$V = \frac{1}{3}\pi\left(8^2\right)(20) - \frac{1}{3}\pi\left(x^2\right)(20-y) = \frac{1280\pi}{3} - \frac{1}{3}\pi\left(x^2\right)(20-y)$$

Since $\tan\theta = \frac{x}{20-y} = 0.4,\ x = 0.4(20-y)$

Thus, $V = \frac{1280\pi}{3} - \frac{1}{3}\pi\left[0.4(20-y)\right]^2(20-y) = \frac{1280\pi}{3} - \frac{0.16}{3}\pi(20-y)^3$

Method 1:

Differentiate w.r.t t,

$\frac{\mathrm{d}V}{\mathrm{d}t} = 0.16\pi(20-y)^2\frac{\mathrm{d}y}{\mathrm{d}t}$

When $y = 10$, $\frac{\mathrm{d}V}{\mathrm{d}t} = 8$,

$8 = 0.16\pi(10)^2\frac{\mathrm{d}y}{\mathrm{d}t}$

$\frac{\mathrm{d}y}{\mathrm{d}t} = \frac{1}{2\pi} = 0.159$ (3.s.f)

$\therefore$ Depth of water increases at 0.159 cm per second.

Method 2:

Differentiate w.r.t y

$\frac{\mathrm{d}V}{\mathrm{d}y} = 0.16\pi(20-y)^2$

$\frac{\mathrm{d}V}{\mathrm{d}t} = \frac{\mathrm{d}V}{\mathrm{d}y}\times\frac{\mathrm{d}y}{\mathrm{d}t} = 0.16\pi(20-y)^2\times\frac{\mathrm{d}y}{\mathrm{d}t}$

When $y = 10$, $\frac{\mathrm{d}V}{\mathrm{d}t} = 8$,

$8 = 0.16\pi(10)^2\frac{\mathrm{d}y}{\mathrm{d}t}$

$\frac{\mathrm{d}y}{\mathrm{d}t} = \frac{1}{2\pi} = 0.159$ (3.s.f)

$\therefore$ Depth of water increases at 0.159 cm per second.

3 RVHS Prelim 9758/2021/02/Q3

A curve C has equation $y^2+xy=-1$, where $y>-1$.

(a) Describe a sequence of transformations that will transform the graph of C onto the graph of $4y^2-2(x-1)y=-1$. [3]

(b) (i) Show that the gradient function of C can be expressed as $\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{-y}{2y+x}$. [2]

(ii) For a non-zero constant a, find the equation of the tangent to the point where $y=a$ in terms of a. [3]

(iii) State the behaviour of the tangent as $a\to 1$. [1]

Solution:

(a) Note that $4y^2-2(x-1)y=-1$ is equivalent to $(2y)^2+2y(1-x)=-1$.

$$\begin{aligned} & y^2+xy=-1 \\ \xrightarrow{\text{A: replace } y \text{ with } 2y} \quad & (2y)^2+x(2y)=-1 \\ & \Rightarrow 4y^2+2xy=-1 \\ \xrightarrow{\text{B: replace } x \text{ with } -x} \quad & 4y^2+2(-x)y=-1 \\ & \Rightarrow 4y^2-2xy=-1 \\ \xrightarrow{\text{C: replace } x \text{ with } x-1} \quad & 4y^2-2(x-1)y=-1 \end{aligned}$$

A: Scale the graph parallel to the y-axis by factor $\frac{1}{2}$

B: Reflect the graph about the y-axis

C: Translate the graph 1 unit in the positive x-direction

OR:

$$\begin{aligned} & y^2+xy=-1 \\ \xrightarrow{\text{A: replace } y \text{ with } 2y} \quad & (2y)^2+x(2y)=-1 \\ & \Rightarrow 4y^2+2xy=-1 \\ \xrightarrow{\text{B: replace } x \text{ with } x+1} \quad & 4y^2+2(x+1)y=-1 \\ \xrightarrow{\text{C: replace } x \text{ with } -x} \quad & 2y^2+2((-x)+1)y=-1 \\ & \Rightarrow 2y^2-2(x-1)y=-1 \end{aligned}$$

A: Scale the graph parallel to the y-axis by factor $\frac{1}{2}$

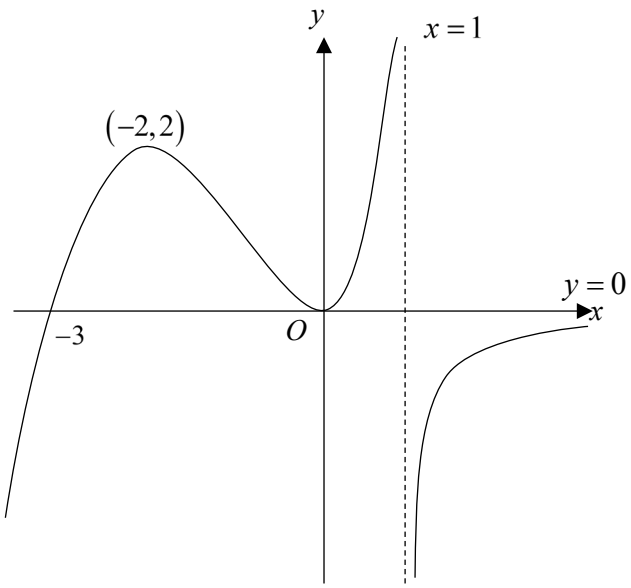
B:Translate the graph 1 unit in the negative x-direction

C: Reflect the graph about the y-axis

	Note: (A) can be applied independently of (B) and (C), eg (B) followed by (C) followed by (A) is another possible sequence.
(b) (i)	$y^2 + xy = -1$ Differentiate w.r.t x, $2y\frac{\mathrm{d}y}{\mathrm{d}x} + x\frac{\mathrm{d}y}{\mathrm{d}x} + y = 0$ $\frac{\mathrm{d}y}{\mathrm{d}x}(2y + x) = -y$ $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{-y}{2y + x}$ (shown)
(b) (ii)	When $y = a$, $a^2 + ax = -1$ $x = -\frac{a^2 + 1}{a}$ $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{-a}{2a - \frac{a^2 + 1}{a}}$ $= \frac{-a^2}{a^2 - 1}$ $= \frac{a^2}{1 - a^2}$ Equation of tangent where $y = a$ is $y - a = \frac{a^2}{1 - a^2}\left(x - \left(-\frac{a^2 + 1}{a}\right)\right)$ $y = \frac{a^2}{1 - a^2}x + \left(a + \frac{a(a^2 + 1)}{1 - a^2}\right)$ $= \frac{a^2}{1 - a^2}x + \frac{a(1 - a^2) + a(1 + a^2)}{1 - a^2}$ $= \frac{a^2}{1 - a^2}x + \frac{2a}{1 - a^2}$
(b) (iii)	As $a \to 1, x \to -2$ and $\frac{\mathrm{d}y}{\mathrm{d}x} \to \infty$. The tangent will tend to the vertical line $x = -2$.

4 EJC Prelim 9758/2021/01/Q4

(a) The graph of $y = \mathrm{f}(x)$ is given below. It has one vertical asymptote at $x = 1$ and one horizontal asymptote at $y = 0$. The graph cuts the x-axis at $x = -3$ and has turning points at $(-2, 2)$ and the origin.



Sketch the graph of $y = \mathrm{f}'(x)$, stating the equations of any asymptotes, and indicating any axial intercepts. [3]

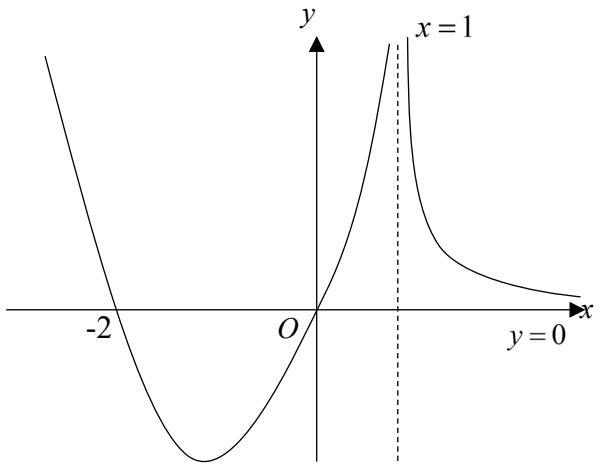
(b) It is given that $\mathrm{g}(x) = x^3 - 3x + p$, where p is an unknown constant.

(i) Given that the graph of $y = \dfrac{1}{\mathrm{g}(x)}$ has a vertical asymptote at $x = 2$, find the value of p. [1]

(ii) Given instead that the graph of $y = \dfrac{1}{\mathrm{g}(x)}$ has a minimum point at $y = \dfrac{1}{5}$, find the value of p. [3]

Solution:

(a)



(b) (i)

If $y = \dfrac{1}{\mathrm{g}(x)}$ has $x = 2$ as a vertical asymptote, then $y = \mathrm{g}(x)$ has an x-intercept at $x = 2$.

So $\mathrm{g}(2) = 0 \Rightarrow p = -2$.

(b) (ii)

Intended method:

If $y = \dfrac{1}{\mathrm{g}(x)}$ has a minimum point at $y = \dfrac{1}{5}$, then $y = \mathrm{g}(x)$ has a maximum point at $y = 5$.

$\mathrm{g}'(x) = 3x^2 - 3 = 3(x+1)(x-1)$ so stationary points of $y = \mathrm{g}(x)$ are at $x = \pm 1$.

$\mathrm{g}''(x) = 6x$ so, by second derivative test,

$x = 1$ is a minimum point while $x = -1$ is a maximum point.

So turning point is at $(-1, 5)$.

$(-1)^3 - 3(-1) + p = 5 \Rightarrow p = 3$.

Alternatively (since question did not state without using a calculator),

If $y = \dfrac{1}{\mathrm{g}(x)}$ has a minimum point at $y = \dfrac{1}{5}$, then $y = \mathrm{g}(x)$ has a maximum point at $y = 5$.

From GC, observe that $y = x^3 - 3x$ has a maximum point at $y = 2$.

Since we need to move the maximum point to $y = 5$, we have $p = 3$.

5 NYJC Prelim 9758/2022/01/Q7

A curve is defined parametrically by

$$x = \frac{at}{t+1},\ y = \frac{at^2}{t+1},$$

where a is a non-zero constant and $t \in \mathbb{R}, t \neq -1$.

(i) Show that the equation of the normal to the curve at the point T with parameter t is given by $t(t+1)(t+2)y + (t+1)x = at(t^3 + 2t^2 + 1)$. [4]

(ii) The normal to the curve at the point $P\left(-a, \frac{1}{2}a\right)$ meets the curve again at point Q. Find the exact coordinates of Q in terms of a. [4]

(iii) Let M be the mid-point of PQ. Find a cartesian equation of the curve traced by M if a varies. [2]

Solution:

(i)

$x = \frac{at}{t+1}$ $\qquad$ $y = \frac{at^2}{t+1}$

$\frac{dx}{dt} = \frac{a(t+1)-at}{(t+1)^2} = \frac{a}{(t+1)^2}$ $\qquad$ $\frac{dy}{dt} = \frac{2at(t+1)-at^2}{(t+1)^2} = \frac{at^2+2at}{(t+1)^2}$

$$\therefore \frac{dy}{dx} = \frac{\frac{at^2+2at}{(t+1)^2}}{\frac{a}{(t+1)^2}} = t^2 + 2t = t(t+2)$$

$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$

Equation of normal at point T is

$$y - \frac{at^2}{t+1} = -\frac{1}{t(t+2)}\left[x - \frac{at}{t+1}\right]$$

$$t(t+2)y - \frac{at^2}{t+1} \times t(t+2) = -x + \frac{at}{t+1}$$

$$t(t+1)(t+2)y - at^3(t+2) = -(t+1)x + at$$

$$t(t+1)(t+2)y + (t+1)x = at(t^3 + 2t^2 + 1) \text{ (shown)}$$

For perpendicular lines with gradients m_1 and m_2: $m_1 m_2 = -1$

(ii)

At $P\left(-a, \frac{1}{2}a\right)$, $\frac{at}{t+1} = -a \Rightarrow t = -\frac{1}{2}$.

When $t = -\frac{1}{2}$, $\frac{dy}{dx} = -\frac{3}{4}$

Equation of normal at P is $y-\frac{1}{2}a=\frac{4}{3}(x+a) \Rightarrow y=\frac{4}{3}x+\frac{11}{6}a$

OR :

Using result in **(i)**, equation of normal at P is $-\frac{3}{8}y+\frac{1}{2}x=-\frac{11}{16}a$

$$\Rightarrow -6y+8x=-11a$$

Given that this normal meets the curve again at Q,

$$-6\left(\frac{at^2}{t+1}\right)+8\left(\frac{at}{t+1}\right)=-11a$$

$$\Rightarrow 6t^2-8t=11(t+1)$$

$$\Rightarrow 6t^2-19t-11=0$$

$$\Rightarrow (2t+1)(3t-11)=0$$

$$\Rightarrow t=-\frac{1}{2} \text{ or } \frac{11}{3} \quad \left(t=-\frac{1}{2} \text{ is rejected as this corresponds to point } P\right)$$

It is not necessary to convert the parametric equations of curve C to cartesian equation.

When $t=\frac{11}{3}$, $x=\frac{a\left(\frac{11}{3}\right)}{\left(\frac{11}{3}\right)+1}=\frac{11}{14}a$, $y=\frac{a\left(\frac{11}{3}\right)^2}{\left(\frac{11}{3}\right)+1}=\frac{121}{42}a$

$\therefore$ The coordinates of Q are $\left(\frac{11}{14}a,\frac{121}{42}a\right)$

(iii)

Mid-point of PQ, M is $\left(-\frac{3}{28}a,\frac{71}{42}a\right)$

i.e $x=-\frac{3}{28}a$ and $y=\frac{71}{42}a \Rightarrow y=-\frac{142}{9}x$

6 JPJC Prelim 9758/2023/01/Q4

The curve C has equation $y = \dfrac{x^2+x+3}{x-2}$.

(i) Without using a calculator, find the set of values that y can take. [4]

(ii) Sketch the graph of C, stating clearly the equations of any asymptotes and the coordinates of any turning points and axial intercepts. [3]

(iii) By drawing a suitable graph on the same diagram in **(ii)**, find the range of values of k, where $k > 0$, such that the equation $(x+1)^2(x-2)^2 + \left(x^2+x+3\right)^2 = k^2\left(x-2\right)^2$ has at least one positive real root, giving your answer correct to 3 decimal places. [3]

Solution:

(i)

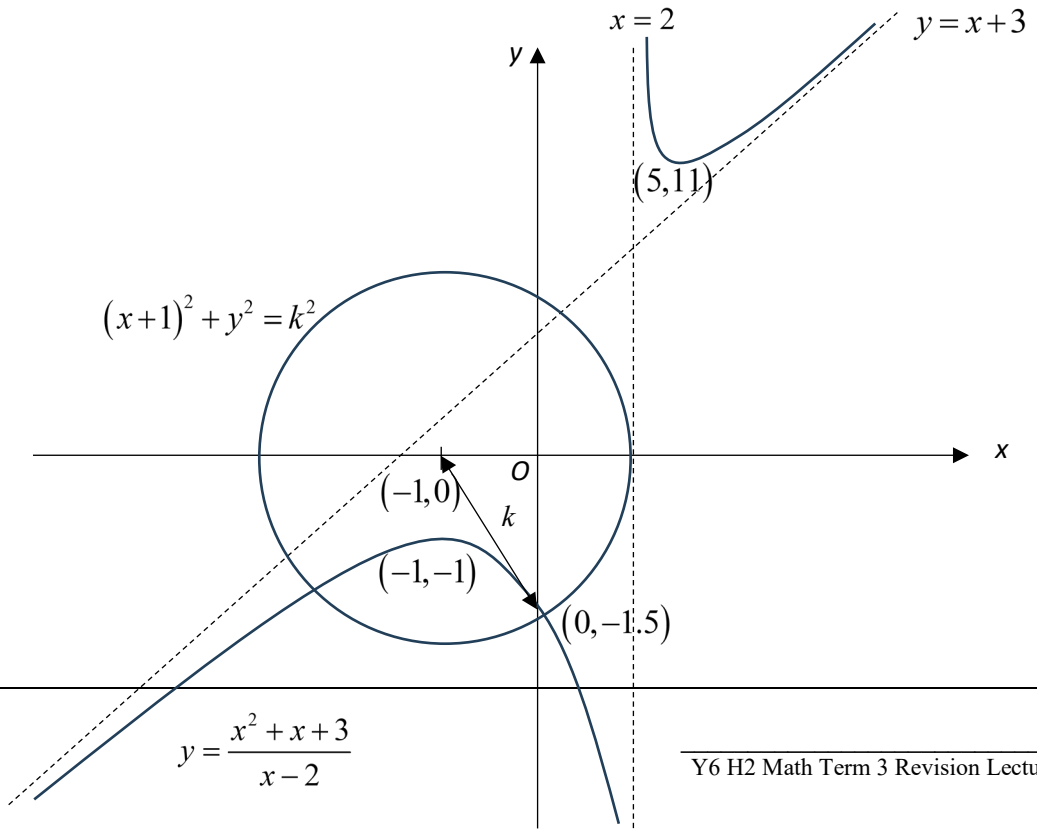
$$y = \frac{x^2+x+3}{x-2}$$
$$yx - 2y = x^2+x+3$$
$$x^2 + (1-y)x + 3 + 2y = 0$$

Since y takes on some value for corresponding value(s) of x, discriminant ≥ 0.

$$(1-y)^2 - 4(1)(3+2y) \geq 0$$
$$y^2 - 10y - 11 \geq 0$$
$$(y+1)(y-11) \geq 0$$
$$y \leq -1 \quad \text{or} \quad y \geq 11$$

Hence, the set of values of y is $\{y \in \mathbb{R} : y \leq -1 \text{ or } y \geq 11\}$.

(ii)



(iii)	$(x+1)^2(x-2)^2+\left(x^2+x+3\right)^2=k^2\left(x-2\right)^2$ $(x+1)^2+\left(\dfrac{x^2+x+3}{x-2}\right)^2=k^2$ $(x+1)^2+y^2=k^2$ For $(x+1)^2(x-2)^2+\left(x^2+x+3\right)^2=k^2\left(x-2\right)^2$ to have at least 1 positive real root, Since $k>0$, $k>\sqrt{\left(-1-0\right)^2+\left(0-(-1.5)\right)^2}$ $k>\sqrt{3.25}$ $k>1.803$ (3 d.p)

7 VJC Prelim 9758/2021/02/Q3

(a) Expand $\dfrac{4+x^2}{\sqrt{4-x}}$ in ascending powers of x, up to and including the term in x^2. [3]

State the set of values of x for which the expansion is valid. [1]

(b) It is given that $y=\ln(1+\sin x)$.

(i) Show that $(1+\sin x)\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}+(\cos x)\dfrac{\mathrm{d}y}{\mathrm{d}x}=-\sin x$. [2]

(ii) By further differentiation of the result in part **(i)**, find the Maclaurin series for y, up to and including the term in x^3. [4]

(iii) Hence deduce the Maclaurin series for $\ln\left(\dfrac{1-\sin x}{2}\right)$, up to and including the term in x^3. [2]

Solution:

(a)

$$\frac{4+x^2}{\sqrt{4-x}}=\left(4+x^2\right)(4-x)^{-\frac{1}{2}}$$

$$=\left(4+x^2\right)\left[4\left(1-\frac{x}{4}\right)\right]^{-\frac{1}{2}}$$

$$=\frac{1}{2}\left(4+x^2\right)\left(1+\left(-\frac{1}{2}\right)\left(-\frac{x}{4}\right)+\frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{2!}\left(-\frac{x}{4}\right)^2+\ldots\right)$$

$$=\frac{1}{2}\left(4+x^2\right)\left(1+\frac{x}{8}+\frac{3}{128}x^2+\ldots\right)$$

$$=\frac{1}{2}\left(4+\frac{1}{2}x+\frac{3}{32}x^2+x^2+\ldots\right)$$

$$=2+\frac{1}{4}x+\frac{35}{64}x^2+\ldots$$

Expansion is valid for

$\left|-\dfrac{x}{4}\right|<1\Rightarrow -4<x<4$

$\therefore\{x\in\mathbb{R}:-4<x<4\}$

(b) (i)

$$y=\ln(1+\sin x)$$

$$\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{\cos x}{1+\sin x}$$

$$(1+\sin x)\frac{\mathrm{d}y}{\mathrm{d}x}=\cos x$$

Differentiate w.r.t x,

$$(1+\sin x)\frac{\mathrm{d}^2y}{\mathrm{d}x^2}+\cos x\frac{\mathrm{d}y}{\mathrm{d}x}=-\sin x \quad \text{(shown)}$$

(ii)

Differentiate w.r.t. x,

$$(1+\sin x)\frac{\mathrm{d}^3y}{\mathrm{d}x^3}+\cos x\frac{\mathrm{d}^2y}{\mathrm{d}x^2}+\cos x\frac{\mathrm{d}^2y}{\mathrm{d}x^2}-\sin x\frac{\mathrm{d}y}{\mathrm{d}x}=-\cos x$$

When $x=0$,

$y=\ln(1+0)=0$

$$\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{\cos 0}{1+\sin 0}=1$$

$$(1+0)\frac{\mathrm{d}^2y}{\mathrm{d}x^2}+1=0$$

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2}=-1$$

$$(1+0)\frac{\mathrm{d}^3y}{\mathrm{d}x^3}+(1)(-1)+(1)(-1)-(0)(1)=-1$$

$$\frac{\mathrm{d}^3y}{\mathrm{d}x^3}=1$$

$$\therefore y=0+(1)x+\frac{(-1)}{2!}x^2+\frac{(1)}{3!}x^3+\ldots$$

$$=x-\frac{1}{2}x^2+\frac{1}{6}x^3+\ldots$$

(iii)

$$\ln\left(\frac{1-\sin x}{2}\right)=\ln(1-\sin x)-\ln 2$$

$$=\ln(1+\sin(-x))-\ln 2$$

$$=-\ln 2+(-x)-\frac{1}{2}(-x)^2+\frac{1}{6}(-x)^3+\ldots \text{ replacing } x \text{ by } -x \text{ in (ii)}$$

$$=-\ln 2-x-\frac{1}{2}x^2-\frac{1}{6}x^3+\ldots$$

8 TJC Prelim 9758/2021/02/Q3

(a) In the triangle ABC, $AC = BC = 1$ and angle $ACB = \left(\frac{1}{2}\pi + \theta\right)$ radians. Given that θ is sufficiently small angle, show that

$$AB \approx \sqrt{2} + \frac{\sqrt{2}}{2}\theta - \frac{\sqrt{2}}{8}\theta^2$$ [5]

(b) Let $\mathrm{f}(x) = \ln(2 - x)$.

(i) Write down the first three non-zero terms of the Maclaurin series for $\mathrm{f}(x)$, where $-2 \le x < 2$.

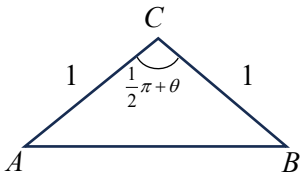
[3]

Denote the answer to part **(i)** by $\mathrm{g}(x)$.

(ii) Find the range of values of x for which the percentage error between the value of $\mathrm{g}(x)$ and the value of $\mathrm{f}(x)$ is within 10%. [2]

Solution:

(a) Using cosine rule,



$$\begin{aligned} AB^2 &= AC^2 + BC^2 - 2(AC)(BC)\cos ACB \\ &= 1 + 1 - 2(1)(1)\cos\left(\frac{1}{2}\pi + \theta\right) \\ &= 2 - 2\left[\cos\left(\frac{1}{2}\pi\right)\cos\theta - \sin\left(\frac{1}{2}\pi\right)\sin\theta\right] \\ &= 2 + 2\sin\theta \end{aligned}$$

When θ is sufficiently small, $AB \approx (2 + 2\theta)^{\frac{1}{2}}$

$$\begin{aligned} &= \left[2(1+\theta)\right]^{\frac{1}{2}} \\ &= \sqrt{2}\left[1 + \frac{1}{2}\theta + \frac{\frac{1}{2}\left(-\frac{1}{2}\right)}{2!}\theta^2 + \ldots\right] \\ &\approx \sqrt{2}\left(1 + \frac{1}{2}\theta - \frac{1}{8}\theta^2\right) \quad \text{(shown)} \end{aligned}$$

(b) (i)

$$\begin{aligned} f(x) &= \ln(2-x) \\ &= \ln\left[2\left(1-\frac{1}{2}x\right)\right] \\ &= \ln 2 + \ln\left(1-\frac{1}{2}x\right) \\ &= \ln 2 + \left[\left(-\frac{1}{2}x\right) - \frac{1}{2}\left(-\frac{1}{2}x\right)^2 + \ldots \right] \\ &\approx \ln 2 - \frac{1}{2}x - \frac{1}{8}x^2 \end{aligned}$$

(b) (ii)

$$g(x) = \ln 2 - \frac{1}{2}x - \frac{1}{8}x^2$$

$$\left|\frac{g(x)-f(x)}{f(x)} \times 100\%\right| < 10\%$$

$$\Rightarrow \quad -0.1 < \frac{g(x)-f(x)}{f(x)} < 0.1$$

Using GC(graphically), $-1.73 < x < 0.737$ (correct to 3 s.f.)